

HJARTASEN, Norway

WMO: 011500

Lat: 66.499N

Lon: 14.954E

Elev: 251

StdP: 98.35

Time Zone: 1.00 (EUC)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.7	-19.1	-24.2	0.4	-20.0	-22.5	0.5	-18.4	10.2	-0.7	8.4	-2.7	1.5	90	0.680

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	24.9	15.5	22.3	14.6	19.9	13.1	16.8	22.8	15.4	20.5	14.2	18.6	2.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.6	10.7	18.4	13.2	9.7	16.8	12.0	9.0	16.0	47.9	23.1	44.0	20.6	40.6	18.6	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.3	5.3	DB	-23.8	27.7	2.5	3.0	-25.6	29.9	-27.0	31.6	-28.5	33.2	-30.3	35.4	(4)
(5)			WB	-24.1	18.1	2.5	2.1	-25.9	19.6	-27.3	20.9	-28.7	22.1	-30.5	23.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-7.6	-6.2	-4.2	0.6	5.6	9.8	13.8	12.1	8.6	2.3	-2.8	-4.6	(6)	
	DBStd	8.53	6.26	6.07	5.19	3.60	3.71	3.07	3.57	2.82	3.24	3.82	5.51	5.62	(7)	
	HDD10.0	3065	547	454	441	282	146	41	4	11	62	239	384	454	(8)	
	HDD18.3	5853	805	687	699	532	394	257	148	194	292	497	634	712	(9)	
	CDD10.0	264	0	0	0	0	10	34	123	76	21	1	0	0	(10)	
	CDD18.3	9	0	0	0	0	0	0	8	1	0	0	0	0	(11)	
	CDH23.3	138	0	0	0	0	6	3	118	11	0	0	0	0	(12)	
	CDH26.7	23	0	0	0	0	1	0	22	0	0	0	0	0	(13)	
(14) Wind	WSAvg	2.1	2.2	2.5	2.4	2.2	2.3	2.2	1.8	1.8	1.9	2.1	2.1	2.2	(14)	
(15) Precipitation	PrecAvg	1254	154	138	124	57	59	74	86	89	101	125	95	145	(15)	
	PrecMax	1498	311	286	259	125	90	122	200	147	228	309	250	335	(16)	
	PrecMin	991	22	17	22	25	32	18	21	39	24	19	10	45	(17)	
	PrecStd	164	92	74	74	31	15	30	40	31	56	68	72	88	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.3	5.5	6.4	12.1	23.5	23.6	29.1	24.8	20.0	12.3	8.3	5.7	(19)	
		MCWB	4.3	3.5	2.8	6.2	12.2	14.2	18.2	15.6	12.9	8.2	6.2	4.0	(20)	
	2%	DB	3.0	4.2	5.1	9.1	18.3	20.3	26.6	22.6	16.7	10.3	7.2	4.4	(21)	
		MCWB	1.4	2.6	2.6	3.8	10.3	12.2	16.7	15.0	11.5	7.5	5.6	2.9	(22)	
	5%	DB	1.4	3.1	3.8	6.9	15.6	18.4	24.2	20.2	15.1	8.8	5.1	3.1	(23)	
		MCWB	0.3	1.4	1.6	2.8	9.0	11.0	15.8	14.1	11.0	6.5	3.5	1.5	(24)	
	10%	DB	0.3	1.8	2.5	5.4	12.2	16.4	21.3	17.9	13.6	7.5	3.3	1.8	(25)	
		MCWB	-0.6	0.4	0.6	2.2	7.4	10.4	14.5	13.0	10.4	5.8	1.9	0.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.3	4.1	4.1	6.3	12.5	14.7	19.9	17.3	14.1	9.2	7.1	4.6	(27)	
		MCDB	5.2	5.1	5.6	11.7	22.3	20.4	26.8	21.5	17.7	11.3	7.9	5.3	(28)	
	2%	WB	1.8	2.9	2.9	4.6	10.7	13.4	17.9	15.8	12.7	7.7	5.5	3.0	(29)	
		MCDB	2.6	4.2	4.5	7.6	17.7	18.5	24.1	20.0	15.7	9.7	7.0	4.2	(30)	
	5%	WB	0.4	1.6	1.8	3.5	9.2	12.2	16.4	14.7	11.6	6.8	3.8	1.7	(31)	
		MCDB	1.2	2.8	3.6	6.0	14.6	16.9	22.7	19.4	14.2	8.5	5.1	2.7	(32)	
	10%	WB	-0.5	0.5	0.7	2.6	7.7	10.9	15.2	13.5	10.6	6.0	2.1	0.7	(33)	
		MCDB	0.3	1.6	2.2	4.8	11.9	15.3	20.3	17.4	13.0	7.5	3.2	1.6	(34)	
(35) Mean Daily Temperature Range	MDBR		5.8	6.7	8.2	8.2	8.4	8.7	9.8	9.4	7.9	5.4	4.9	5.5	(35)	
	5% DB	MCDBR	5.2	5.6	6.7	9.7	13.9	13.9	15.9	14.3	10.7	5.9	4.9	5.5	(36)	
		MCWBR	4.5	4.5	4.8	5.8	7.1	6.8	7.0	7.5	6.2	4.2	4.0	4.3	(37)	
	5% WB	MCDBR	5.2	5.6	6.0	8.0	13.0	12.0	13.3	12.6	9.3	5.0	4.8	5.4	(38)	
		MCWBR	4.7	4.7	4.6	4.9	6.8	6.1	6.3	7.0	5.7	3.7	3.9	4.5	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.192	0.259	0.297	0.323	0.333	0.333	0.344	0.330	0.304	0.272	0.197	0.197	(40)	
	taud		1.911	2.102	2.134	2.159	2.244	2.370	2.422	2.463	2.509	2.272	1.959	1.923	(41)	
	Ebn at Noon		209	583	749	829	863	875	851	828	770	572	189	15	(42)	
	Edh at Noon		16	53	87	111	113	103	95	83	62	44	14	3	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.06	0.44	1.75	3.66	4.89	4.93	4.47	3.38	1.78	0.62	0.14	0.01	(44)	
	RadStd		0.01	0.08	0.25	0.30	0.27	0.44	0.53	0.36	0.20	0.11	0.01	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (27 neighbors)	+0.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page